

Executable post-tracking refinement path

All refiners receive the same frozen upstream tracklets

1 Saved tracklets

fixed boxes + classes
local IDs + frame support
ByteTrack / OC-SORT /
Deep OC-SORT

2 Tracklet evidence

first / middle / last crops
mean 128-D ReID
descriptor + endpoints

3 Candidate graph

$0 < \Delta t \leq 30$ frames
 $d_{xy} \leq 55$ px
 $d_{cos} \leq 0.30$

4 Reciprocal gate

mutual best successor
deterministic tie breaking
unmatched is allowed

5 Component guard

reject component merges
with frame overlap
preserve every input box

6 Refined output

accepted ID remaps
unresolved stay unchanged
no interpolation

Evaluation-only audit: GT identities are read after links are frozen for HOTA / AssA / IDF1 / IDSW and link TP / FP